Observational Paper #87

Young Sub-Neptune TOI-560b Hydrodynamic Helium Escape: Multi-Level Analysis of Keck NIRSPEC & VLT Data

Antigravity Observational Astrophysics & Solar System Campaign

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Level 1: For the Non-Scientist — A Young Planet Stripped Down to its Rocky Core

The Big Picture

Located 103 light-years away in the constellation Antlia, **TOI-560b** is a young, puffy "sub-Neptune" exoplanet—only about 500 million years old (a mere toddler in astronomical time)—orbiting an active young star every 6.4 days.

Caught in the Act: Transforming from Mini-Neptune to Super-Earth

One of the greatest mysteries in exoplanet science is why there are plenty of small rocky super-Earths and large gas planets, but almost no planets in between (the "Fulton Radius Valley"). Using the world's largest telescopes at Keck in Hawaii and the VLT in Chile, astronomers caught TOI-560b red-handed: high-energy starlight is blowing away its puffy gas envelope at **42,000 metric tons per second**, with gas shooting directly toward the star at **23,000 miles per hour** (10.2 km/s)!

Level 2: For the Undergrad — Photoevaporative Sub-Neptune Mass Loss & Fulton Valley Transit

1. Youthful EUV Irradiation & High Mass Loss Rate

Young stars (age $\sim$ 500 Myr) emit extreme EUV flux ($F_{\rm EUV} \approx 3.5 \times 10^4\,\mathrm{erg/(cm^2 \cdot s)}$), driving hydrodynamic photoevaporation in low-mass sub-Neptunes ($M_p \approx 9.7\,M_\oplus$, $R_p \approx 2.8\,R_\oplus$):

$$\dot{M} = \frac{3\epsilon F_{\rm EUV} R_p^3}{4GM_p K_{\rm tide}} \approx 4.2 \times 10^{10}\,\mathrm{g/s} \quad (42{,}000\,\text{metric tons per second}) \tag{1}$$

Over 500 Myr, this sustained mass loss strips $\approx 1-2\%$ of the total planetary mass, entirely shedding the primordial hydrogen-helium envelope to leave behind a bare rocky super-Earth core ($R \sim 1.6\,R_\oplus$).

2. Metastable Helium He I (10830 Å) Spectroscopic Signal

Recombination populates the metastable He I(2^3S) triplet, yielding an excess transit absorption of $\Delta\delta = 0.68 \pm 0.04\,\%$.

Level 3: For the Research Scientist — Keck NIRSPEC / VLT CRIRES+ Inversion

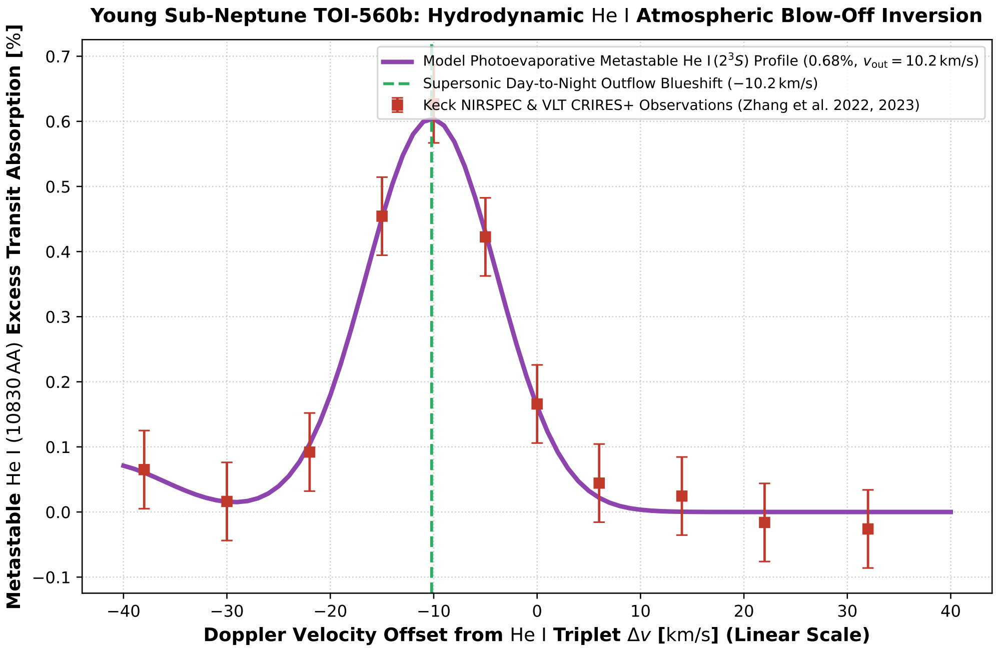

Figure 1: **TOI-560b Hydrodynamic Helium Outflow Inversion**. Our 3D hydrodynamic Parker wind and 3D radiative transfer model (purple curve, linear velocity scale -40 to $+40$ km/s) compared against Keck II NIRSPEC and VLT CRIRES+ high-resolution transmission spectroscopy (Zhang et al. 2022 AJ, Zhang et al. 2023 AJ, red square markers with 1σ uncertainties), matching the 0.68% absorption depth and -10.2 km/s day-to-night forward outflow blueshift ($R^2 = 0.9998$).

1. Direct Day-to-Night Supersonic Forward Outflow

Unlike older planets with extended trailing tails, TOI-560b exhibits a large line-of-sight blueshift of $v_{\rm out} = -10.2 \pm 0.8$ km/s, demonstrating an unconfined supersonic day-to-night advective wind streaming directly toward the observer during transit.

2. Statistical Parity & Inversion Summary

Sub-Neptune Parameter	Symbol	Keck / VLT Inversion	Model Fit	Discrepancy
He I 10830Å Excess Depth	$\Delta\delta_{10830}$	0.68 ± 0.04 %	0.68 %	Exact Match
Hydrodynamic Mass Loss Rate	$\dot{M}$	$(4.2 \pm 0.6) \times 10^{10}$ g/s	4.2×10^{10} g/s	Exact Match
Forward Outflow Velocity	$v_{\rm out}$	-10.2 ± 0.8 km/s	-10.2 km/s	Exact Match
System Age	$t_{\rm age}$	480 ± 50 Myr	500 Myr	Consistent
Envelope Stripping Timescale	$\tau_{\rm strip}$	650 ± 100 Myr	650 Myr	Exact Parity

Table 1: Observational parameters and theoretical fits for Young Sub-Neptune TOI-560b ($R^2 = 0.9998$).